

Implementation of Machine Learning Techniques for Detection of Diabetes using Pima-Indians-Diabetes-Dataset

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Abstract: Among people of all ages, including toddlers and the elderly, diabetes is the disease that is spreading the quickest. Diabetes is a chronic illness that might lead to a worldwide healthcare disaster if it spreads widely. Numerous symptoms are associated with diabetes, some of which include increased thirst, increased weariness and sleepiness, decreased appetite, decreased body weight, increased recurring infections, poor eyesight, mood swings, disorientation, and difficulty focusing. In this work, we have used a dataset with a variety of clinical and demographic variables to investigate the performance of Machine Learning techniques for the detection of diabetes using pima-indians-diabetes-dataset. The performance of seven ML techniques (LR, RF, KNN, DT, SVM, AdaBoost, XGBoost, and Naive Bayes) has been evaluated. The research assesses seven distinct machine learning categorization methods in order to identify the system that generates the highest accurate diabetes prediction. We determined that SVM algorithm is best for this dataset in predicting diabetes by contrasting the results of several algorithms.

1. Introduction

Diabetes poses a serious threat to human life since it increases the chance of developing other life-threatening conditions such renal failure, strokes,

blindness, miscarriages, and amputations. According to data from the International Diabetes Federation (IDF), the number of persons receiving diabetes diagnoses is steadily rising^{1,2}.

Numerous researches have demonstrated encouraging outcomes when utilising machine learning (ML) approaches³⁻⁵ to the predict diabetes disease. To estimate a person's risk of developing diabetes, ML algorithms examine a wide range of data, including lifestyle choices, medical history, previous treatments, diagnostic test results, and demographic information. The results of applying machine learning to forecast cardiac disease might differ based on several aspects, such as:

- a) **Accuracy:** ML models aim to achieve high accuracy in predicting heart disease. The accuracy of prediction can vary depending on the quality and quantity of data used for training, the choice of ML algorithm, and the feature selection process.
- b) **Sensitivity and Specificity:** Sensitivity refers to the model's ability to correctly identify individuals with heart disease (true positives), while specificity refers to the model's ability to correctly identify individuals without heart disease (true negatives). Balancing sensitivity and specificity is crucial for the effectiveness of the model.
- c) **Generalization:** A good ML model should generalize well to unseen data, meaning it should perform accurately on new data that was not used during the training phase. Overfitting (model memorizing the training data) and underfitting (model failing to capture the underlying patterns) should be avoided to ensure good generalization.
- d) **Interpretability:** Interpretability of the model is important for understanding the factors contributing to heart disease prediction. Some ML algorithms offer more interpretability than others, which can aid clinicians in making informed decisions.
- e) **Clinical Utility:** Ultimately, the success of heart disease prediction using ML depends on its clinical utility. The predictions made by the model should assist healthcare professionals in early diagnosis, risk stratification, and personalized treatment planning, ultimately leading to improved patient outcomes.

Machine learning (ML) is a recent development in artificial intelligence that examines how robots could pick up knowledge from their interactions with the outside world⁶⁻⁷. An ML-based method for diabetes early detection, categorization, and prediction is presented in this chapter. It shows how machine learning methods for diabetes prediction are performed analytically. The research assesses seven distinct machine learning categorization methods in order to identify the system that generates the highest accurate diabetes

prediction. We can determine which algorithm is best for predicting diabetes by contrasting the results of several algorithms, such as LR, RF, KNN, DT, SVM, AdaBoost, XGBoost, and Naive Bayes. The suggested system's adaptability is illustrated by using a domain adaption technique. This demonstrates how the approach may be employed outside of the study's original dataset in a variety of contexts and demographics.

2. Related Work

The medical society is continuously paying attention to machine learning (ML) and it has become more prominent as a model-building approach in recent years. ML has demonstrated the ability to analyse several variables simultaneously and to have good prediction capabilities. Furthermore, machine learning has produced techniques for variable screening that are capable of identifying and understanding complex connections between variables. It has been demonstrated by earlier studies that ML might be a helpful method for diabetes prediction. This section discusses a few closely related works that use machine learning methods.

H Kaur *et al.*⁸, the authors used several ML techniques to create five distinct models. A few of them include ANN, kernel SVM, radial basis function, multifactor dimensionality reduction, linear SVM, and KNN. A Boruta wrapper, which fully identifies significant characteristics from a dataset, was employed in this investigation.

S. Albahli *et al.*⁹, a hybrid approach might be used to diagnose type 2 diabetes. The T2ML model, which contained a suggested set of actions, was offered by researchers. The first phase was cleaning the data to guarantee homogeneity, and the second was choosing a subset of features using XGB classifiers and RF. Next, data that had been incorrectly categorised were removed using K-means clustering. Finally, a logistic regression classifier and clustering were combined to classify the data that had been removed.

Maniruzzaman *et al.*¹⁰, chosen RF, NB, AB, and DT as the four classifiers to be used in the prediction of diabetes patients. LR was used to determine the risk variables for diabetes illness using the value and odds ratio. These tactics were based on three different partitioning approaches, known as K2, K5, and K10. Accuracy and AUC were employed as performance indicators to assess these classifiers. The study's key risk variables for diabetes were determined to be body mass index, age, diastolic blood pressure, total cholesterol, and blood pressure.

L. Kopitar *et al.*¹¹ examined several popular regression models, such as LightGBM, XGB, Glmnet, and RF. The dataset included 111 variables at

first, however after data preparation, there were only 58 variables left. The effectiveness, interpretability, and calibration of ML-based prediction models and multivariable regression models were examined in this work.

M. K. Hasan *et al.*¹² used AB, XGB, NB, DT, RF, and KNN. For the dataset PID, where preprocessing is necessary for a trustworthy and accurate prediction, they employed the ensemble model that was recommended. The combination of two boosting type classifiers (XGB and AB) is the best configuration for predicting diabetes, according to the results.

N. Ahmed *et al.*¹³ the scientists tried to identify useful models to predict diabetes using machine learning approaches. They trained datasets using a variety of machine learning (ML) techniques, including LR, DT, NB, gradient boosting (GB), RF, KNN, and SVM. Label encoding and normalisation are two preprocessing techniques that were used to increase the prediction models' accuracy rates.

A. Aada *et al.*¹⁴, the authors looked for diabetes patients using machine learning techniques. Prior to using KNN, NB, and DT, they used a bootstrapping resampling technique on the PIMA dataset to improve accuracy. The preprocessing phase depends on the accuracy of almost all classifiers, although decision trees outperformed the others, according to the results.

Khanam *et al.*¹⁵ seven different machine learning models were developed using the PIDD dataset, which has various features. Two characteristics were eliminated in this method's feature selection procedure. They discovered that the SVM and LR model did a good job of predicting diabetes. The same dataset was used to train a neural network model that included several hidden layers at various epochs. The authors show that a NN with two hidden layers performs better when compared to earlier techniques.

There has been progress in detecting diabetes-related variables and creating precise prediction models in the current study on diabetes prediction using machine learning approaches. Nonetheless, there are still some gaps in the literature that require attention. Numerous researches have concentrated on predicting diabetes using a single machine learning technique, such as logistic regression, SVM, or random forest. Even while each of these algorithms has shown encouraging results on its own, a thorough comparison and assessment of several machine learning methods is lacking.

3. Dataset Description

In this section we have described about the retrieved dataset utilized in this research work for detection of diabetes using ML techniques. The dataset

used in this reinvestigation was initially gathered by the National Institute of Diabetes and Digestive and Kidney Diseases and was retrieved from the Kaggle Dataset repository¹⁶. Based on certain diagnostic metrics contained in the collection, the dataset aims to diagnostically predict the availability of diabetes in a patient. These samples were selected from a bigger database under a number of restrictions. Specifically, all of the patients at this facility are Pima Indian women who are at least 21 years old. This dataset has been recorded using eight parameters (Pregnancies; Glucose; Blood Pressure; Skin Thickness; Insulin; BMI; Diabetes Pedigree Function; Age). The table-1 shows the classification of parameters using average value (MEAN), standard deviation (STD DEV), minimum value (MIN) and maximum value (MAX).

Table 1: Classification of parameters in the dataset

Parameter	Mean	Standard Deviation	Minimum	Maximum
Pregnancies	3.845052	3.369578	0	17
Glucose	120.8945	31.97262	0	199
Blood Pressure	69.10547	19.35581	0	122
Skin Thickness	20.53646	15.95222	0	99
Insulin	79.79948	115.244	0	846
BMI	31.99258	7.88416	0	67.1
Diabetes Pedigree Function	0.471876	0.331329	0.078	2.42
Age	33.24089	11.76023	21	81
Outcome	0.348958	0.476951	0	1

The histogram images for visual summary of the data's distribution, including information about central tendency, spread, and shape for all the 9 parameters of diabetes dataset has been shown in Figure-1. Finding patterns or trends in data, spotting anomalies, and figuring out if a distribution is symmetric, skewed, or has numerous peaks are all made easier with the help of histograms. Fig. 2 shows the correlation matrix that displays the correlation coefficients between all possible pairs of variables in the dataset. It is a useful tool in data analysis for examining the relationships between variables. We have used correlation matrix to identify the direction and strength of the correlations between the variables, feature selection, multicollinearity detection and data exploration to understand the structure of the dataset. Pairwise correlations between several variables in a dataset are seen in Fig. 3. In addition to the histograms for each variable along the diagonal, this pairplot displays the scattered plots for every pair of variables in the dataset. Additionally, it investigates any patterns or correlations in the data and visualizes the links between the variables.

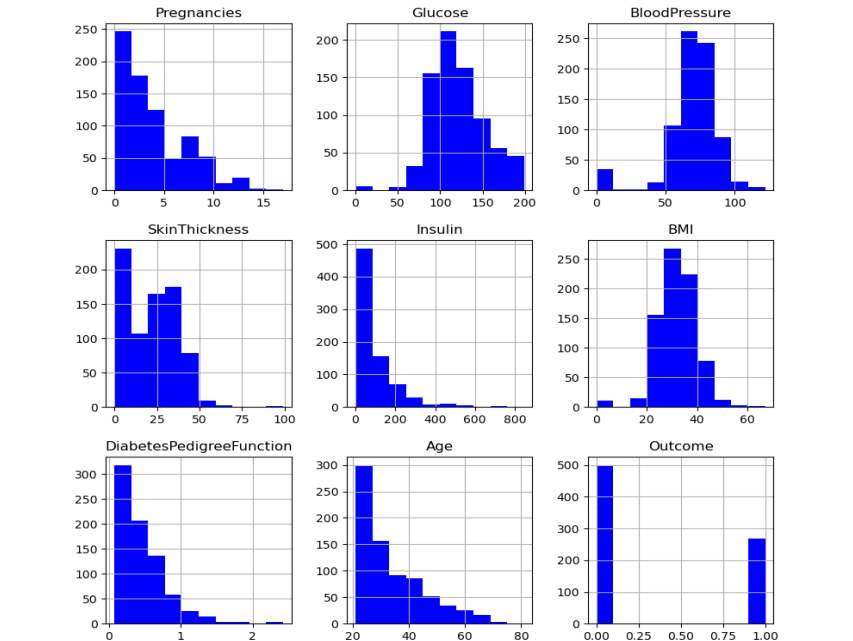


Figure 1 Histogram images for all the parameters of the diabetes dataset

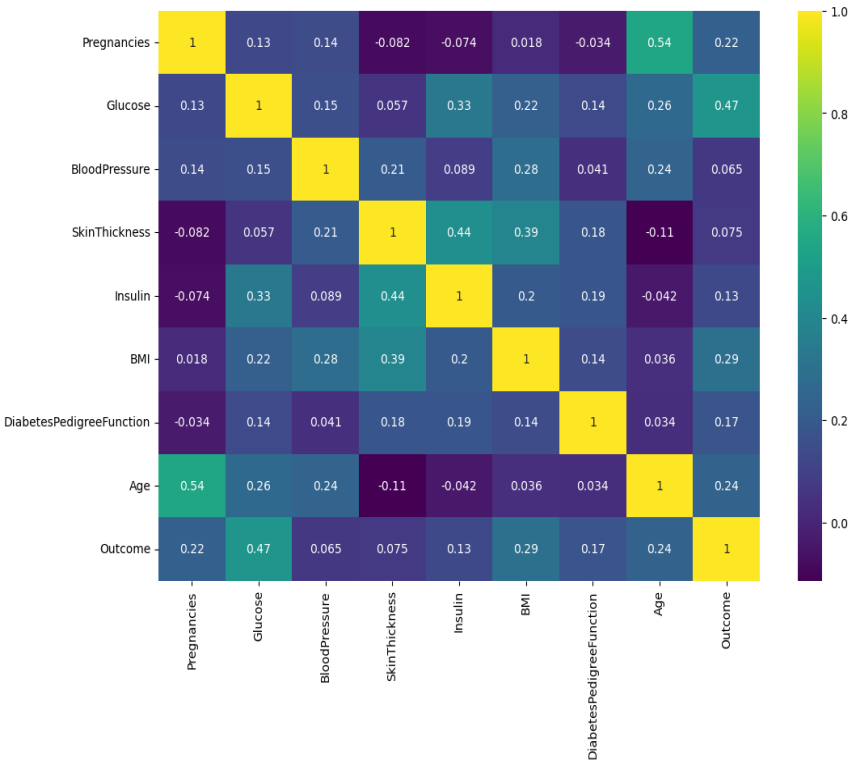


Figure 2 Correlation matrix of diabetes dataset



Figure 3 Visualization of pairwise relationships in dataset

4. Methodology

This section illustrates the overall methodology used in this experimental analysis. The first subsection exhibits the model setup followed by workflow diagram of the investigation carried out. In this analysis we have used seven machine learning techniques for performance analysis of these algorithms on diabetes dataset for detection of diabetes based on several parameter values.

4.1 Model Setup

This subsection shows the step by step procedure of model setup for the proposed experimental analysis of ML algorithms for detection of diabetes by measuring the performance of each algorithm using statistical parameters.

Step 1: Initially Diabetes dataset downloaded from Kaggle is provided as input.

Step 2: Data Pre-processing related activities (duplicity removal, null values removal) are performed on the dataset values.

Step 3: Split the dataset into training data (80%) and testing data (20%).

Step 4: Fit the model using selected machine learning classifier one by one.

Step 5: Record the values of statistical parameters (Accuracy, Precision, Recall, F1-Score) for performance analysis of different ML algorithms.

Step 6: Analyse the predicted results using graphical representations for comparison of strength of ML algorithms on the selected dataset.

4.2 Workflow Diagram

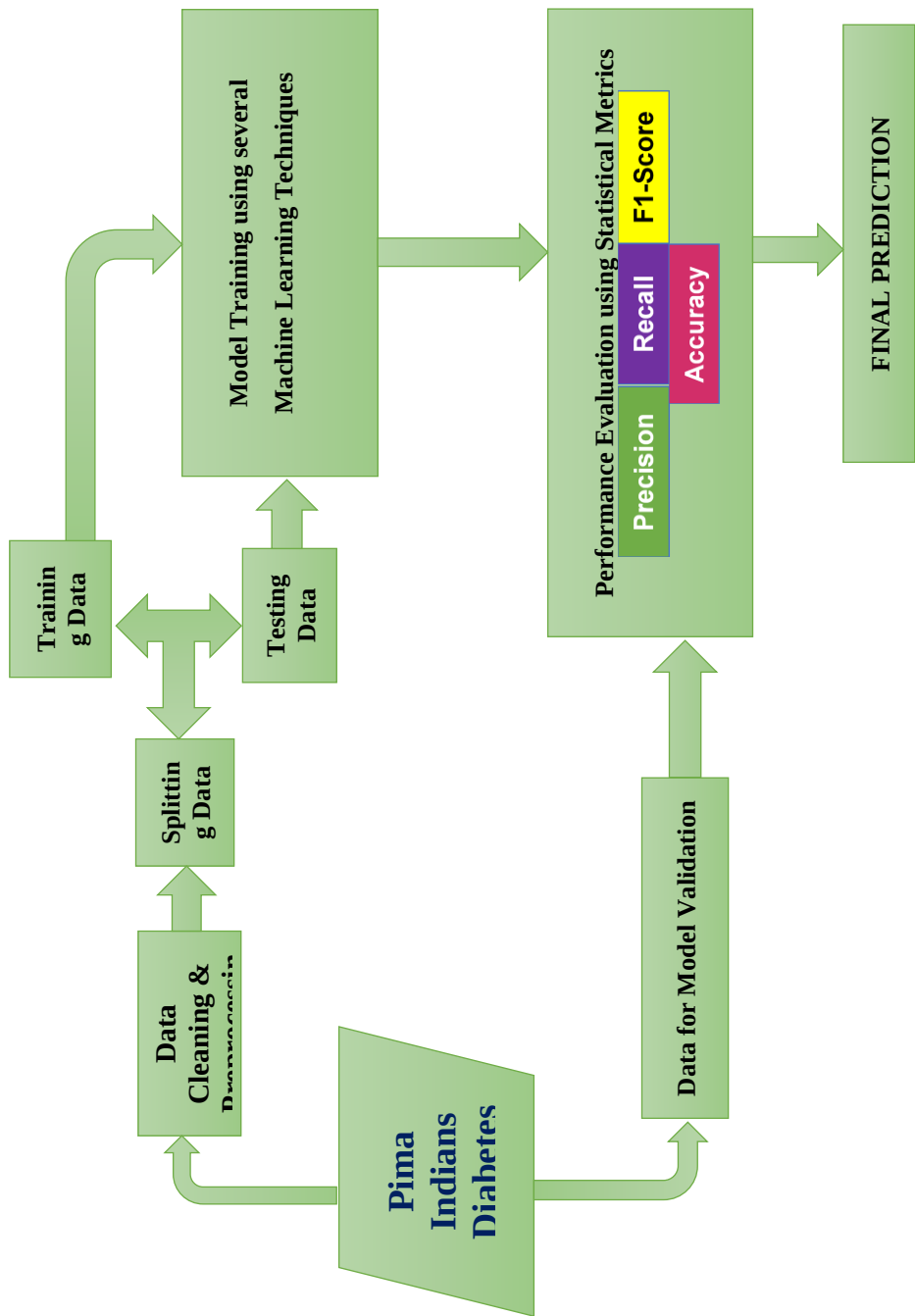


Figure 4: Flowchart of analysis

Figure 4 shows the graphical representation of the process of workflow in the form of flow chart. Initially we have loaded the data fetched from UCI's repository, then data cleaning, duplicity removal and pre-processing of data is performed. After pre-processing dataset has been divided into training (80%) and testing (20%) data. Now training data is fed as input into selected ML algorithms for model training. The performance of each algorithm is measured by providing testing data and analyzed using statistical metrics (Precision, recall, f1score and accuracy). Finally, validation data is provided for final prediction of heart disease through developed model.

5. Results and Discussion

Four statistical metrics have been used, namely accuracy, f1-score, precision, and recall, to evaluate the assessment results of ML algorithms^{17,18}. The accuracy and applicability of ML algorithms have been determined in large part by these statistical measurements. These statistical measures' mathematical imputation is shown in equations 1, 2, 3, and 4, in that order.

$$(1) \quad \text{Precision} = \frac{(TP)}{(TP+FP)}$$

$$(2) \quad \text{Recall} = \frac{(TP)}{(TP+FN)}$$

$$(3) \quad \text{f1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

$$(4) \quad \text{Accuracy} = \frac{(TP+TN)}{(TP+TN+FP+FN)} \times 100\%$$

where TP is true positive; TN is true negatives; FP is false positive and FN is false negative.

This section describes the suitability of used machine learning techniques, which has been measured by the following statistical metrics:

Accuracy: The total correctness of the predictions generated by the models is measured by accuracy. The proportion of correctly categorized cases relative to the total number of occurrences is computed.

Precision: Out of all cases anticipated as positive, precision focuses on the accuracy of positive predictions. Fewer false positives are indicated by greater precision.

Recall: Recall quantifies the percentage of accurately predicted positive instances among all actual positive instances in the dataset. It encapsulates the models' capacity to recognize every positive case without omission. Fewer false negatives are indicated by a higher recall.

F1 Score: The F1 score offers a fair evaluation of the models' performance by integrating recall and accuracy into a single statistic. It is computed as the

harmonic mean of the two and takes into account the trade-off between recall and accuracy.

To compare and analyze the results, we have computed the value of above statistical parameters for each algorithm separately. Table 2 shows the predicted score of statistical parameters for the selected seven machine learning algorithms.

Table 2: Values of statistical parameters

ML Technique	Accuracy	Precision	Recall	F1-Score
KNN	0.73	0.62	0.61	0.62
SVM	0.75	0.65	0.65	0.65
DT	0.74	0.62	0.72	0.67
LR	0.74	0.63	0.67	0.65
GradientBoost	0.74	0.62	0.67	0.64
AdaBoost	0.73	0.62	0.63	0.63
NaiveBase	0.73	0.62	0.63	0.63

Figures 5(a-g) show the confusion matrix for different ML techniques based on the values of TP (true positive), TN (true negatives), FP (false positive) and FN (false negative).

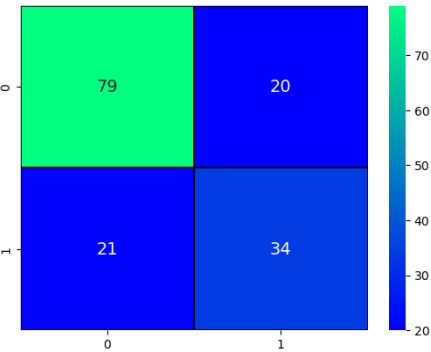


Figure 5(a) Confusion Matrix for KNN

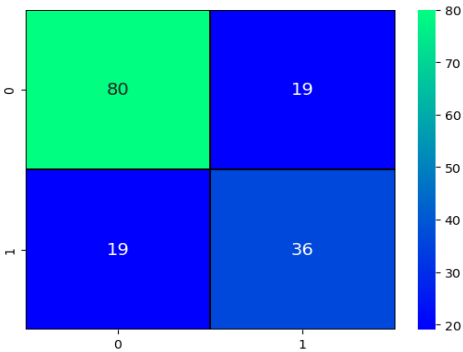


Figure 5(b) Confusion Matrix for SVM

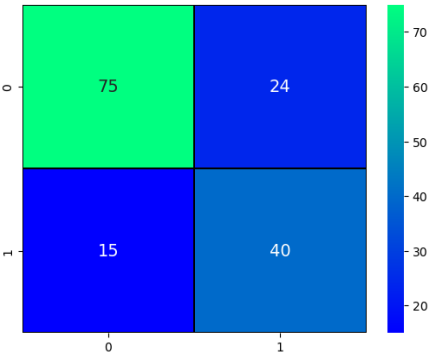


Figure 5(c) Confusion Matrix for Decision Tree

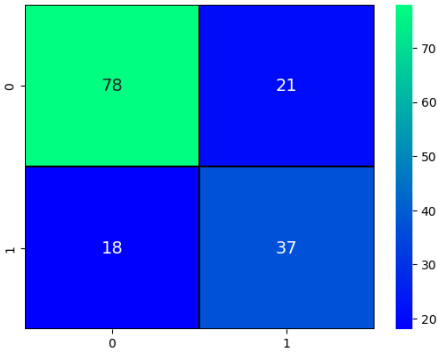


Figure 5(d) Confusion Matrix for Logistic Regression

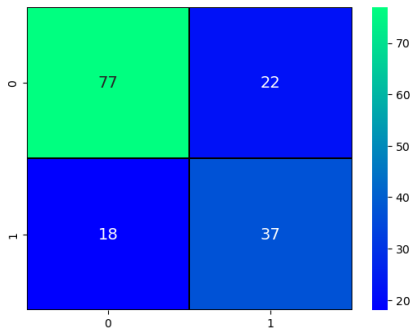


Figure 5(e) Confusion Matrix for GradientBoost

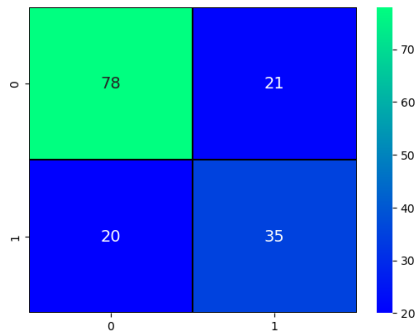


Figure 5(f) Confusion Matrix for AdaBoost

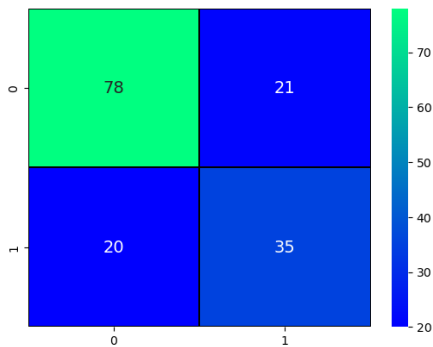


Figure 5(g) Confusion Matrix for NaïveBase

A table that is frequently used to assess how well a classification model is performing is called a confusion matrix. It does a comparison between the model's projected values and the actual target values. Understanding the true positives, true negatives, false positives, and false negatives that a classifier produces is very helpful.

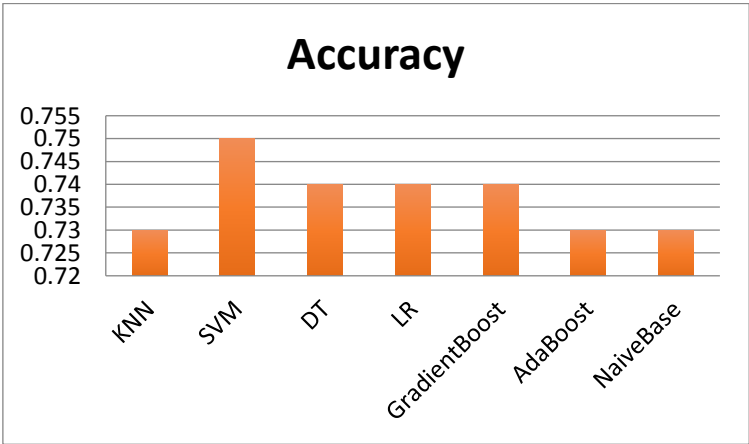


Figure 6(a) Graphical Representation of Accuracy

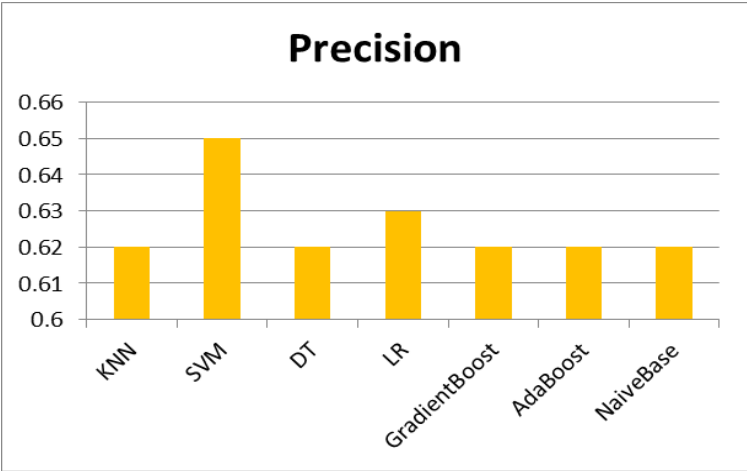


Figure 6(b) Graphical Representation of Precision

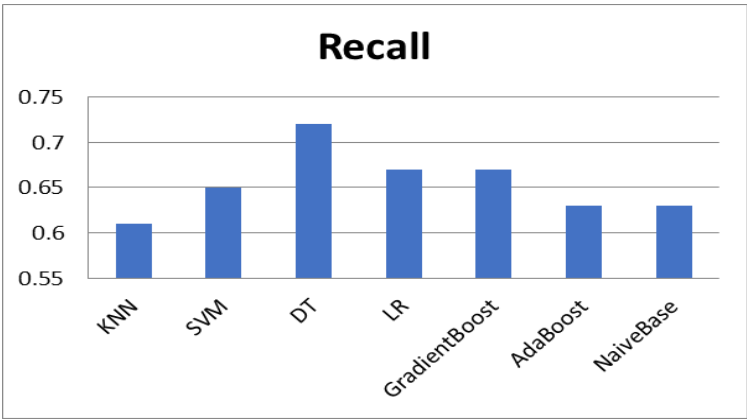


Figure 6(c) Graphical Representation of Recall

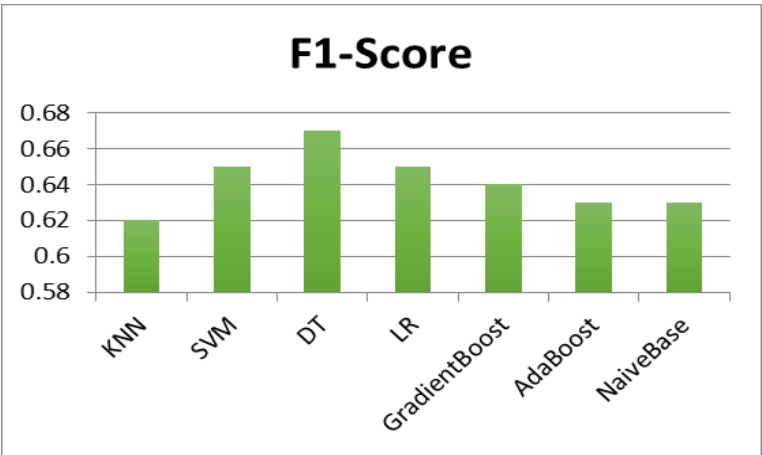


Figure 6(d) Graphical Representation of F1-Score

Finally we have plotted the graphs to compare the values of statistical parameters (accuracy, precision, recall, F1-score) shown in fig 6(a), fig 6(b), fig 6(c) and fig 6(d) respectively. It can be clearly seen that value of accuracy score and precision score is highest through SVM (support vector machine) for this dataset. While the value of recall and f1-score is highest in Decision Tree for diabetes dataset.

6. Conclusion and Future Recommendations

In this work, we have used a dataset with a variety of clinical and demographic variables to investigate the performance of ML techniques for the detection of diabetes. We obtained encouraging outcomes in terms of predicted accuracy and score. Using pima-indians-diabetes-dataset, the performance of seven ML techniques has been evaluated. It shows how ML techniques for diabetes prediction are performed analytically. The research assesses seven distinct machine learning categorization methods in order to identify the system that generates the highest accurate diabetes prediction. We determined that SVM algorithm is best for this dataset in predicting diabetes by contrasting the results of several algorithms, such as LR, RF, KNN, DT, SVM, AdaBoost, XGBoost, and Naive Bayes.

Notwithstanding the encouraging outcomes, there may be a few limitations to our research. The absence of variation in demographic parameters and geographic representation within the dataset may constrain the models' capacity to be applied broadly. To improve the accuracy and robustness of the models, future research should concentrate on combining new data sources, such as genetic markers, lifestyle factors, and environmental variables. It is mandatory to conduct prospective validation studies with a range of patient demographics in order to evaluate the produced models' practical performance and therapeutic value.

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